

LAST MOUNTAIN, SK, Canada

WMO: 715560

Lat: 51.415N

Lon: 105.243W

Elev: 497

StdP: 95.50

Time Zone: -6.00 (W06)

Period: 96-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -33.1	(c) -30.0	(d) -36.6	(e) 0.1	(f) -33.1	(g) -33.3	(h) 0.2	(i) -30.0	(j) 12.4	(k) -7.5	(l) 11.4	(m) -7.7	(n) 2.9	(o) 190	(p) 0.714

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.7	29.7	19.0	27.8	18.2	26.1	17.2	21.3	26.6	20.0	25.1	18.9	23.8	5.2	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.5	15.2	24.0	18.2	13.9	22.4	17.0	12.9	21.3	64.4	26.6	59.7	25.3	55.8	23.8	26.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
Min	Max		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
11.1	9.8	8.8	-37.3	33.5	3.1	1.8	-39.5	34.8	-41.3	35.8	-43.0	36.8	-45.2	38.1		
			-37.3	23.8	3.1	1.2	-39.5	24.7	-41.3	25.4	-43.1	26.0	-45.3	26.9		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	2.5	-14.5	-13.4	-6.6	3.5	10.5	15.3	18.4	17.2	11.9	3.9	-4.9	-12.3	(6)
	DBStd	13.31	8.48	8.03	7.90	5.79	4.67	3.45	3.17	3.57	4.47	4.89	6.92	8.07	(7)
	HDD10.0	3551	760	654	516	203	52	2	0	1	30	196	447	692	(8)
	HDD18.3	5865	1018	887	774	444	244	103	39	65	196	447	697	950	(9)
	CDD10.0	813	0	0	0	8	67	161	260	222	88	7	0	0	(10)
	CDD18.3	86	0	0	0	0	2	11	41	29	4	0	0	0	(11)
	CDH23.3	1156	0	0	0	3	61	138	462	388	101	2	0	0	(12)
	CDH26.7	292	0	0	0	1	9	24	125	108	24	0	0	0	(13)
	WSAvg	4.2	4.3	4.1	4.3	4.7	4.8	4.2	3.9	3.8	4.2	4.3	4.2	4.0	(14)
	PrecAvg	403	16	10	13	28	43	82	67	53	37	25	14	16	(15)
(16) Precipitation	PrecMax	762	41	18	33	65	145	179	143	235	142	68	50	40	(16)
	PrecMin	239	1	1	1	0	9	15	21	9	0	0	1	3	(17)
	PrecStd	123	9	5	9	21	27	41	34	49	28	18	11	9	(18)
	PrecStd	123	9	5	9	21	27	41	34	49	28	18	11	9	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	4.0	4.1	12.6	22.2	27.9	29.6	32.5	32.6	29.5	22.3	12.2	5.7	(19)
		MCWB	2.0	2.0	6.9	11.0	14.6	18.5	20.8	19.7	16.9	12.7	7.1	2.7	(20)
	2%	DB	2.3	1.8	8.6	19.1	25.1	26.7	29.7	29.5	26.4	17.9	8.2	3.0	(21)
		MCWB	0.9	0.6	4.6	9.7	13.9	17.0	20.2	18.6	15.8	10.5	4.7	1.3	(22)
	5%	DB	0.4	0.3	5.6	16.1	22.6	24.7	27.8	27.4	23.4	14.5	5.7	0.9	(23)
		MCWB	-0.6	-0.5	3.1	8.3	12.6	16.1	19.3	18.1	14.7	8.8	3.3	-0.3	(24)
	10%	DB	-2.2	-1.8	3.0	13.2	20.2	23.0	25.9	25.4	20.6	11.8	3.6	-1.5	(25)
		MCWB	-2.8	-2.5	1.6	7.1	11.5	15.1	18.5	17.1	13.3	7.7	2.1	-2.4	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	2.3	2.4	7.4	12.5	16.8	21.3	24.0	21.9	18.8	13.9	7.6	3.0	(27)
		MCDB	3.6	3.7	11.9	19.9	23.1	26.1	28.5	28.1	25.9	20.1	11.3	5.4	(28)
	2%	WB	1.0	0.8	5.0	10.6	15.1	19.2	22.0	20.6	17.1	11.4	5.1	1.4	(29)
		MCDB	2.2	1.7	8.1	17.0	22.0	23.8	27.1	26.5	23.6	16.5	7.5	3.0	(30)
	5%	WB	-0.6	-0.5	3.2	9.2	13.8	17.6	20.7	19.3	15.6	9.6	3.5	-0.3	(31)
		MCDB	0.5	0.3	5.3	15.0	21.0	22.3	25.7	24.9	21.7	13.6	5.3	0.9	(32)
	10%	WB	-2.9	-2.5	1.7	7.6	12.5	16.4	19.5	18.2	14.2	8.1	2.0	-2.3	(33)
		MCDB	-2.3	-1.8	3.0	12.5	18.7	20.9	24.2	23.4	19.7	11.6	3.5	-1.4	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	10.8	10.9	10.8	12.5	13.7	11.8	12.7	13.6	13.4	11.6	9.5	9.7	(35)
		MCDBR	11.7	10.6	12.2	17.7	17.9	15.2	15.7	17.0	18.4	16.5	11.8	11.2	(36)
		MCWBR	10.7	9.4	8.6	9.2	8.1	7.2	7.5	7.5	8.9	9.6	8.4	9.5	(37)
	5% WB	MCDBR	11.7	10.2	11.8	16.4	15.3	12.6	13.9	14.4	16.4	15.4	11.2	11.1	(38)
		MCWBR	10.6	9.1	8.6	9.0	7.4	6.7	7.6	7.2	8.2	9.3	8.2	9.5	(39)
(40) Clear-Sky Solar Irradiance	taub		0.251	0.270	0.286	0.328	0.343	0.353	0.360	0.366	0.318	0.285	0.252	0.237	(40)
	taud		2.311	2.269	2.398	2.396	2.400	2.416	2.398	2.376	2.492	2.546	2.488	2.365	(41)
	Ebn at Noon		784	863	922	913	911	901	887	863	876	849	796	752	(42)
	Edh at Noon		62	86	93	106	111	110	111	107	85	66	52	50	(43)
(44) All-Sky Solar Radiation	RadAvg		1.21	2.27	3.73	4.82	5.89	6.03	6.33	5.11	3.78	2.20	1.12	0.85	(44)
	RadStd		0.12	0.16	0.33	0.40	0.50	0.42	0.37	0.36	0.30	0.30	0.16	0.09	(45)

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (15 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page